

GigaDevice Semiconductor Inc.

GD32A712AEVB

Arm[®] Cortex[®]-M7 32-bit MCU

用户指南

0.0 版本

(2025 年 7 月)

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1. 简介

GD32A712AEVB 评估板使用 GD32A712AIT3 作为主控制器。评估板使用 GD-Link Type-C 接口或者 DC-12V 连接器提供电源。提供包括扩展引脚在内的及 Reset, User KEY, LED, ADC, DAC, CAN, LIN, I2S, I2C, QSPI, SPI 和 UART 转 RS232、USB 接口等外设资源。更多关于开发板的资料可以查看 GD32A712AEVB-LQFP176-V1.1 原理图。

2. 功能引脚分配

表 2-1. 引脚分配

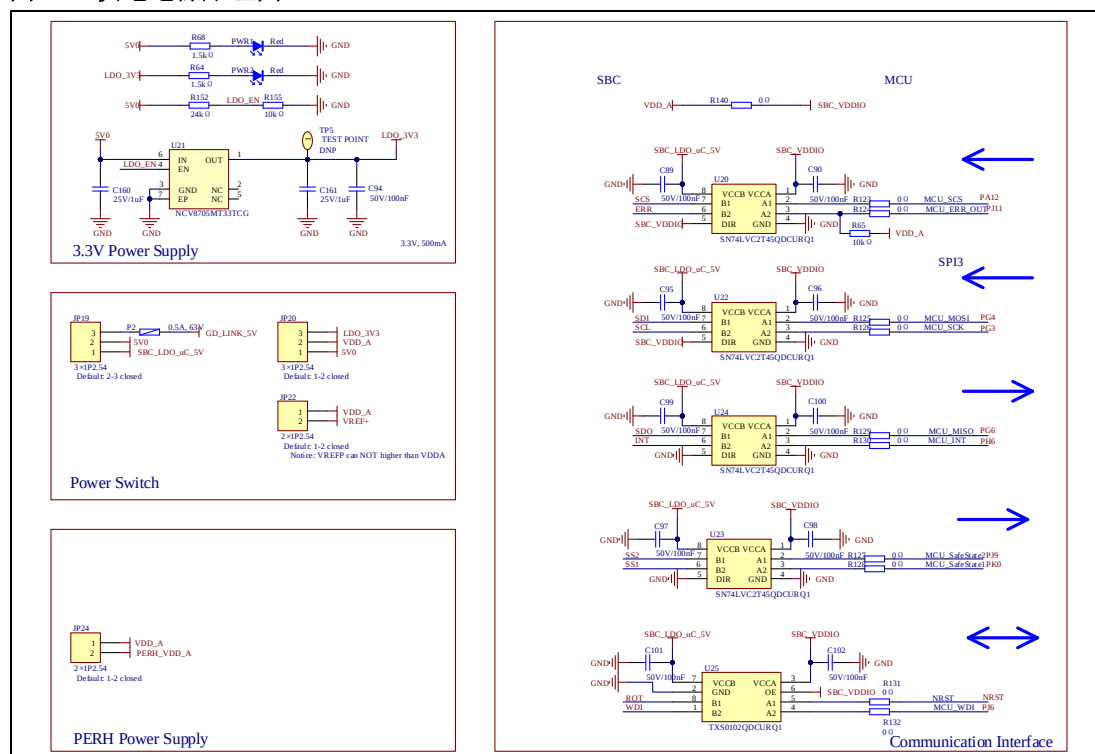
功能	引脚	描述
LED	PE13	LED1
	PE12	LED2
	PE11	LED3
	PE10	LED4
RESET		RESET
USERKEY	PC13	KEY1
	PL3	KEY2
	PL4	KEY3
	PL5	KEY4
ADC	PD11	ADC012_IN9
DAC	PI9	DAC_OUT0
CAN	PH12	CAN0_RX
	PH10	CAN0_TX
	PM3	CAN2RX
	PM2	CAN2TX
	PB12	CAN4RX
	PH11	CAN4TX
	PE14	CAN5RX
	PE15	CAN5TX
LIN	PG0	LIN0_RX
	PF14	LIN0_TX
	PE8	LIN1_RX
	PE7	LIN1_TX
	PD14	LIN2_RX
	PD13	LIN2_TX
	PF13	LIN3_RX
	PJ4	LIN3_TX
I2C	PB2	I2C1_SCL
	PB0	I2C1_SDA
I2S	PB4	I2S_SD

功能	引脚	描述
	PM4	I2S_CK
	PM5	I2S_WS
	PH9	I2S_MCK
QSPI	PH3	QSPI_DQS
	PA4	QSPI_CLK
	PA3	QSPI_IO0
	PH2	QSPI_IO1
	PH5	QSPI_IO2
	PA2	QSPI_IO3
	PA1	QSPI_NSS
	PH4	QSPI_N_RST
SPI	PE6	SPI_SCK
	PH13	SPI_NSS0
	PH14	SPI_MOSI
	PE5	SPI_MISO
	PA8	SPI_IO2
	PM0	SPI_IO3
UART（使用 LINFlexD 的 UART 模式）	PH7	LIN1_TX
	PH8	LIN1_RX
	PJ8	LIN3_TX
	PJ7	LIN3_RX

3. 硬件设计概述

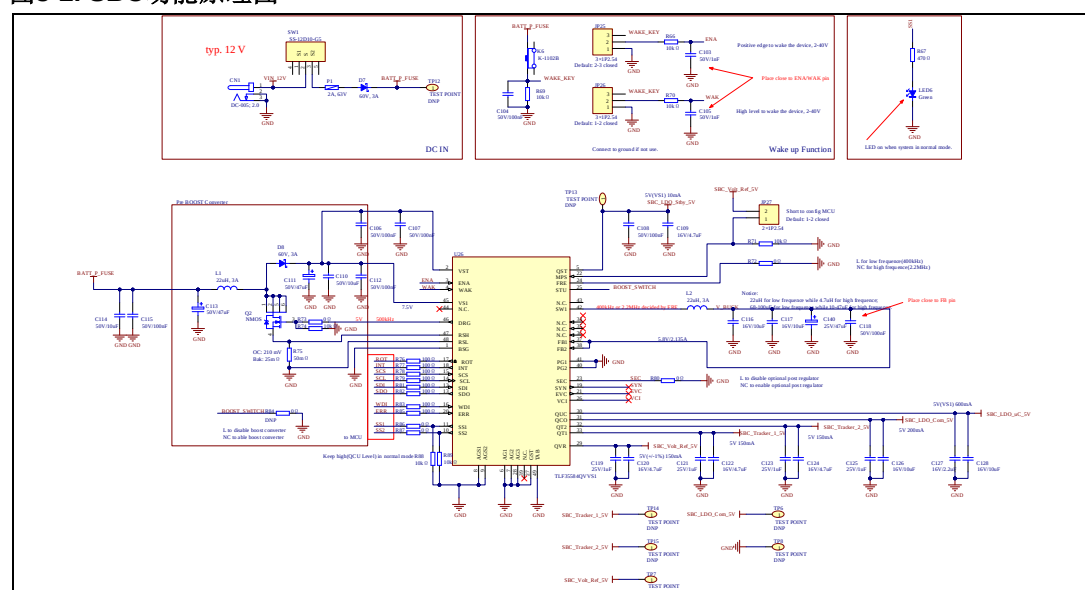
3.1. 供电电源

图3-1. 供电电源原理图



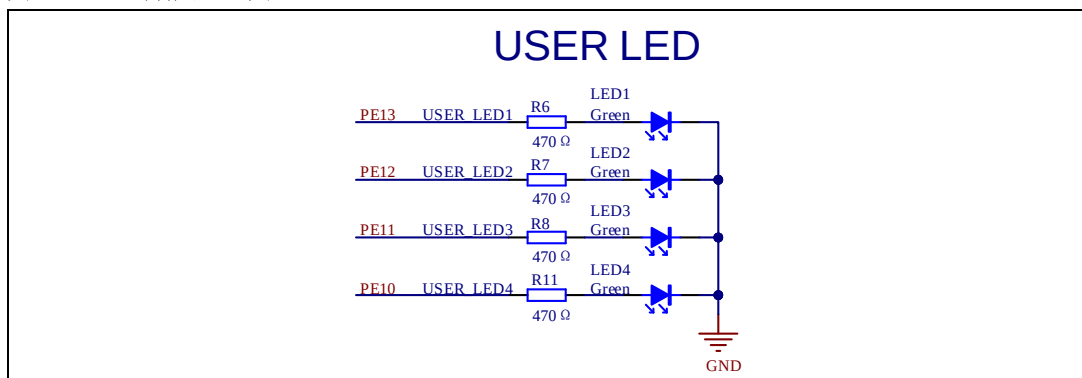
3.2. SBC

图3-2. SBC功能原理图



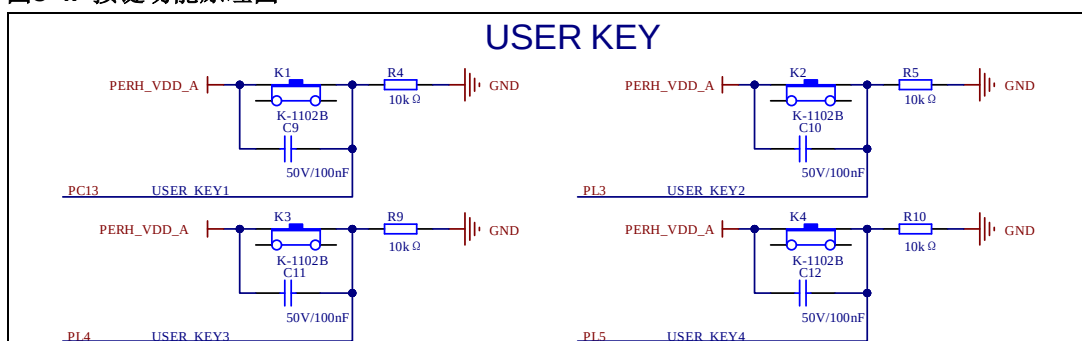
3.3. LED 指示灯

图3-3. LED功能原理图



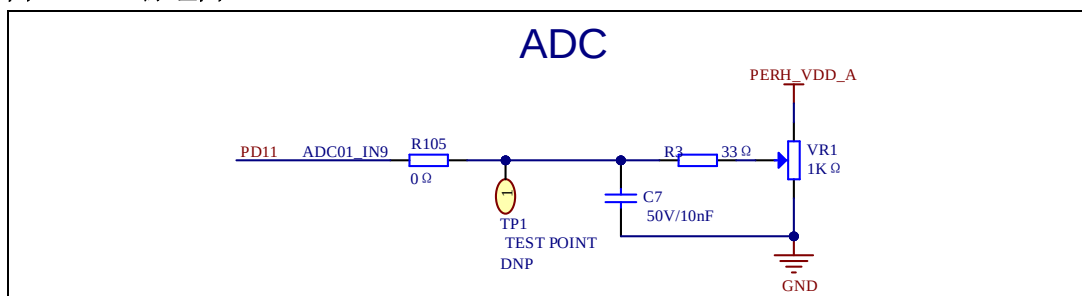
3.4. 按键

图3-4. 按键功能原理图



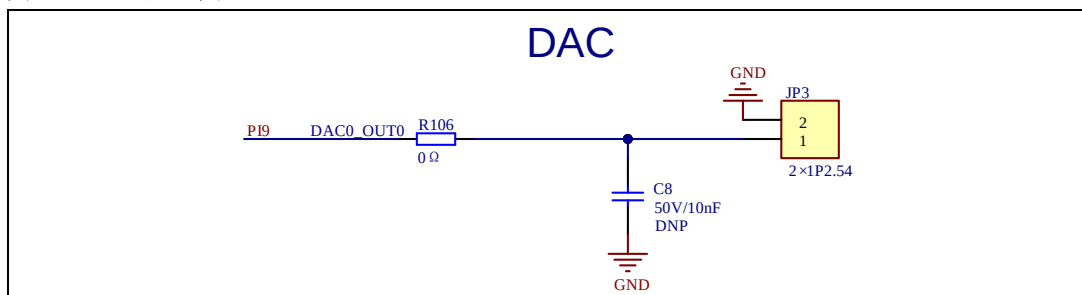
3.5. ADC

图3-5. ADC原理图



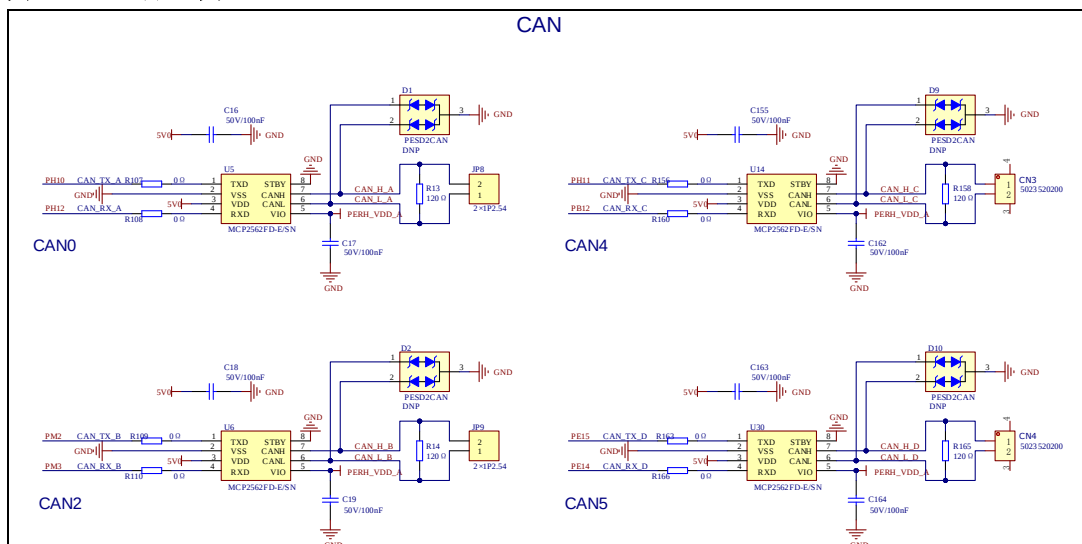
3.6. DAC

图3-6. DAC原理图



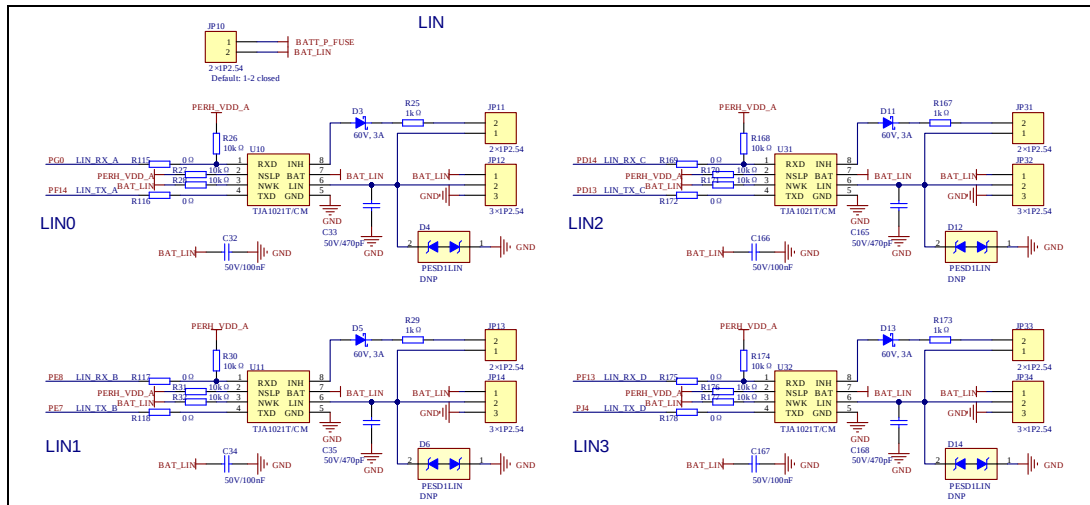
3.7. CAN

图3-7. CAN原理图



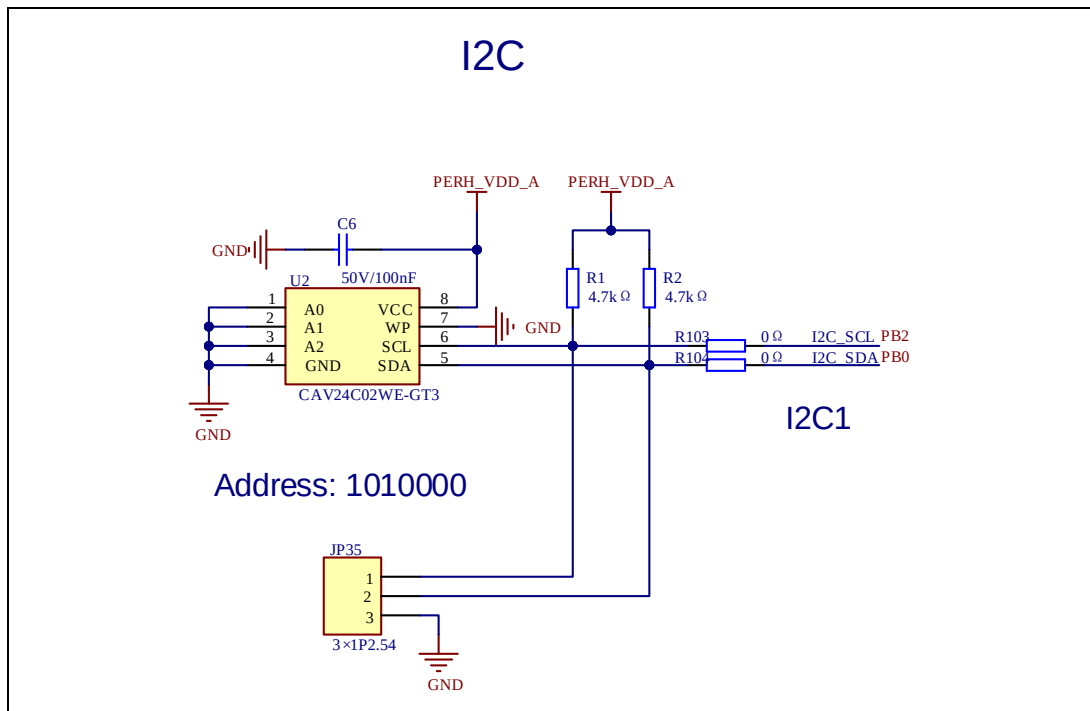
3.8. LINFlexD LIN

图3-8. LIN原理图



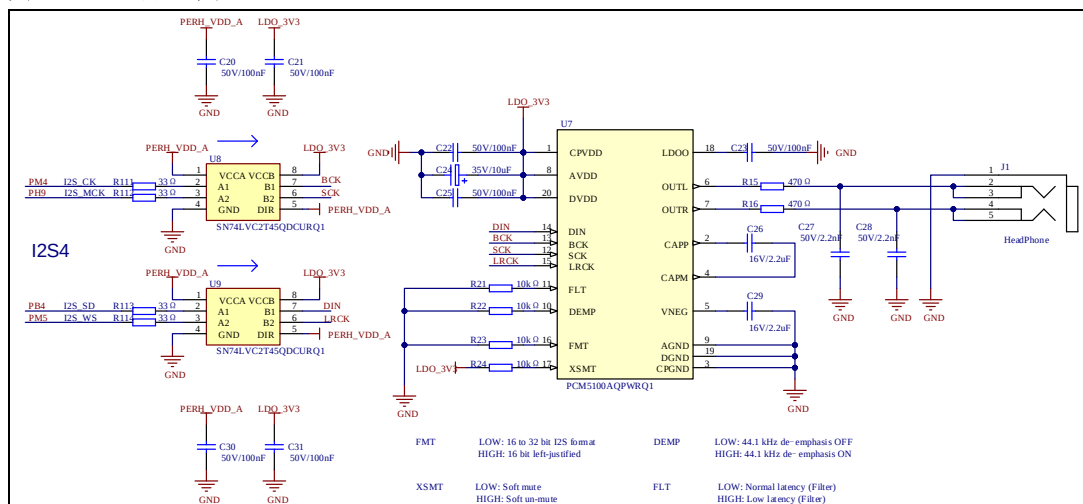
3.9. I2C

图3-9. I2C原理图



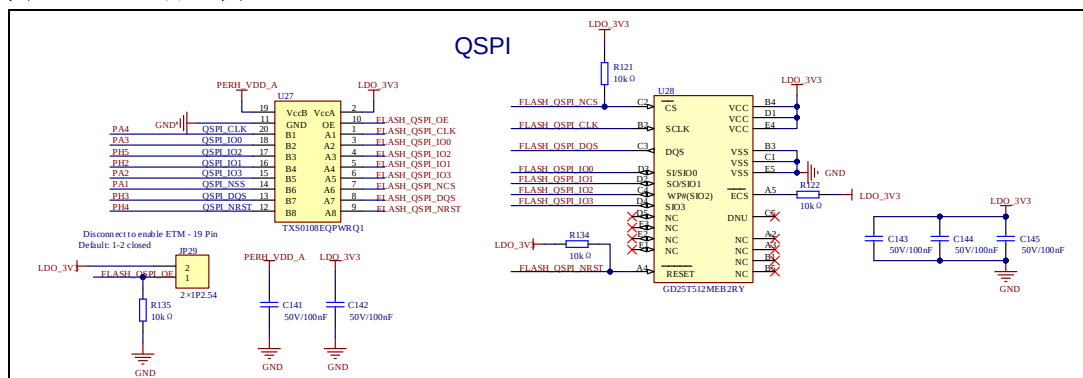
3.10. I2S

图3-10. I2S原理图



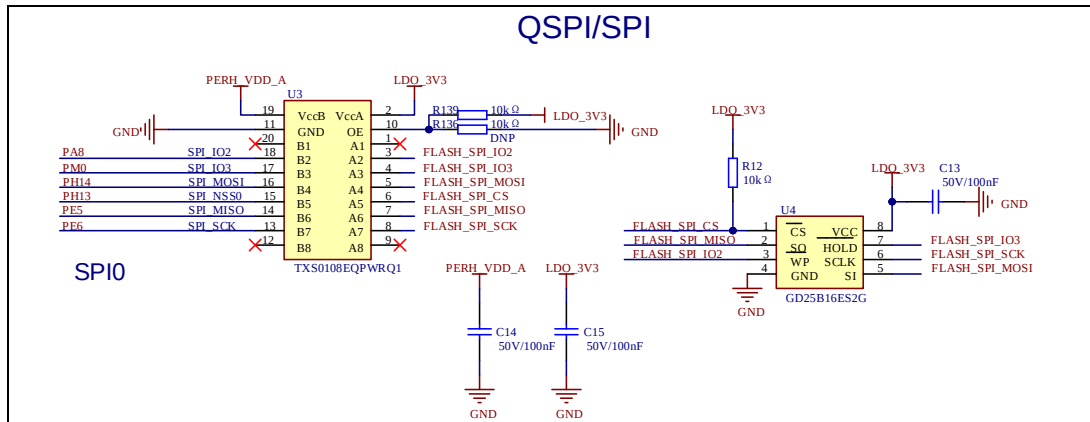
3.11. QSPI

图3-11. OSPI原理图



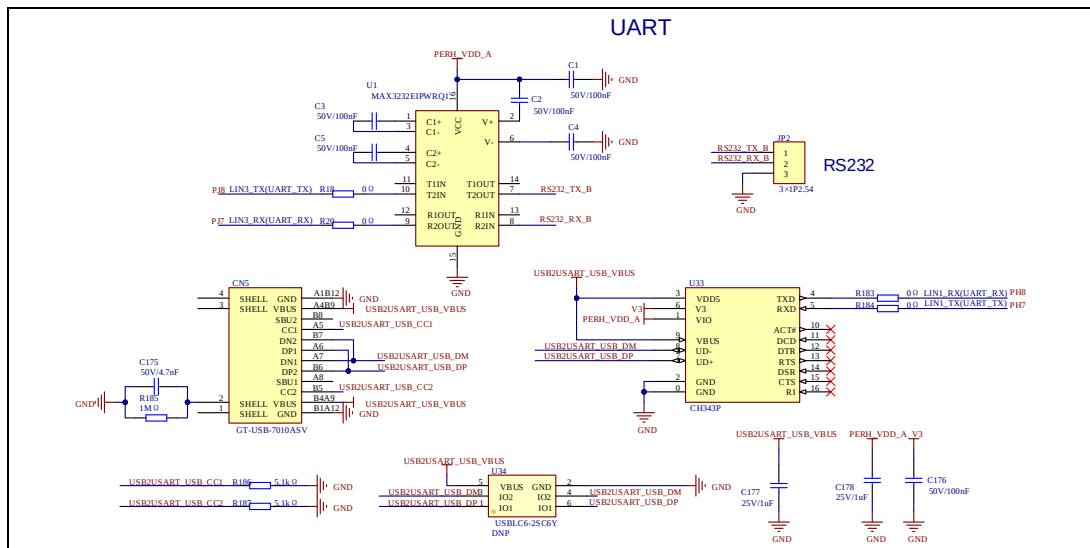
3.12. SPI flash

图3-12. SPI flash原理图



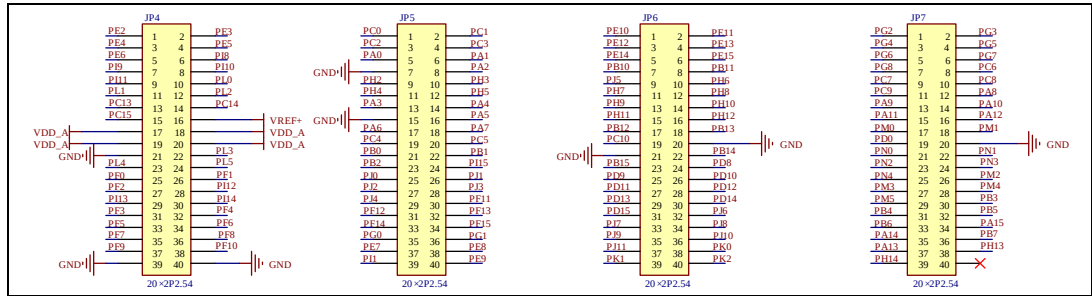
3.13. LINFlexD UART

图3-13.UART原理图



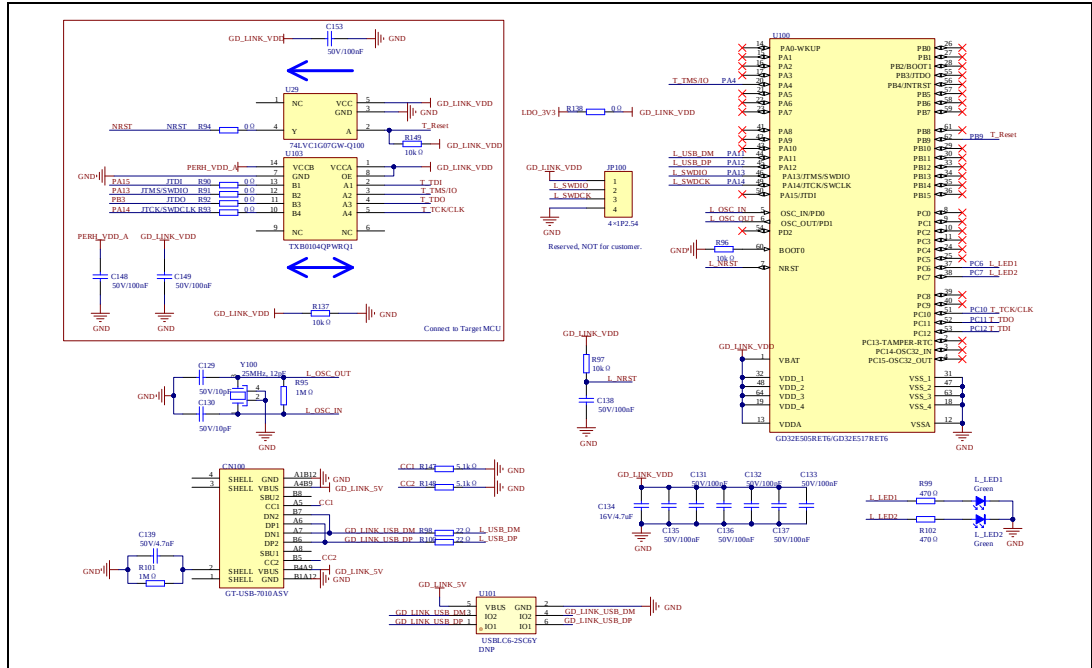
3.14. Extension

图3-14. Extension原理图



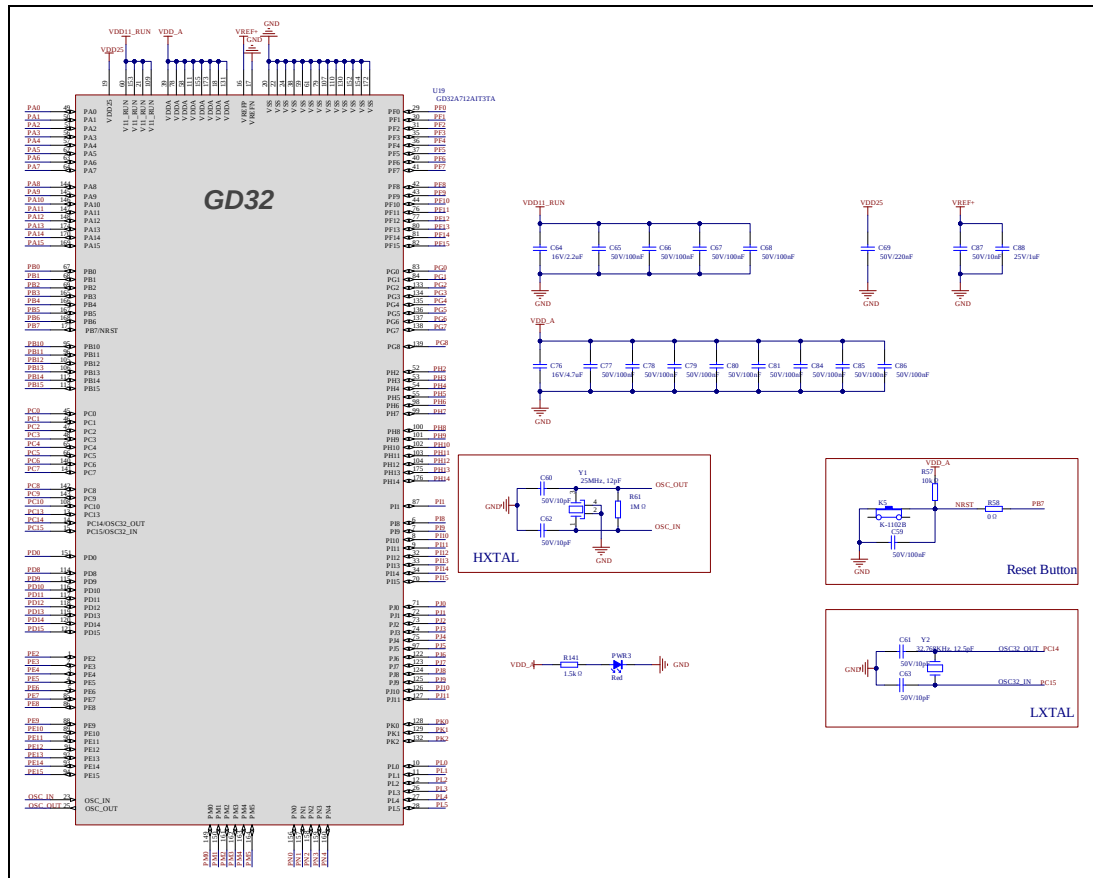
3.15. GD-Link

图3-15. GD-Link原理图



3.16. MCU

图3-16. MCU原理图



4. 版本历史

表 4-1. 版本历史

版本号	说明	日期
0.0	初稿发布	2025 年 7 月 30 日

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